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Obtaining Grain Density of Carbonate Rocks: Core Plugs and Powder Samples

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Abstract

Rock matrix grain density is a basic measurement of routine core analysis used as an input for log quality control and interpretation. It has been reported repeatedly in the literature that laboratory measured grain density of carbonate rocks is systematically lower than the commonly defaulted values. Three hypotheses were formulated as to the root causes of this observation; the cores may not have been fully cleaned and dried, the cores could contain other minerals other than pure carbonates, and there may be presence of isolated/unconnected porosity. The objective of this study is to design an experimental study try to resolve this challenge in routine core analysis.

In this study, we measured the basic rock properties, including porosity and grain density, of 24 outcrop carbonates core samples. The measured porosity and grain density of the samples ranged between 15%-33% and 2.670- 2.830 g/cm^3, respectively. To investigate the effect of drying, samples were subjected to extensive high temperature vacuum drying before measurement of grain density. To evaluate the effect of isolated pores, thin sections and scanning electron microscope (SEM) images were acquired and analyzed for porosity types. Finally, the samples were crushed to powder and powder grain density and mineralogy by using X-ray diffraction (XRD) and Fourier transform infrared spectroscopy (FTIR) measurements were performed.

Results indicate that drying the samples at up to 190°C did not change the grain density. This illustrates that the difference between the measured grain density and that often defaulted is not due to incomplete drying. Mineralogical analysis using XRD and FTIR confirmed the presence of other minerals with lower grain densities, such as quartz and clays. Synthetic grain densities calculated from mineralogy results are higher than those measured from core plugs but still fall short of the expected often defaulted values. Thus, rock mineral impurities may only play part of the role. Finally, thin section and SEM images showed the presence of inaccessible isolated pores. If exists, it will lower the matrix density. The results and findings of this study should help to resolve a long-standing issue in core analysis and lead to a better understanding of why the measured grain density of carbonate rocks tends to be lower than textbook values, attributing to rock mineral impurities and existence of isolated pores.

Introduction

Grain density, or matrix grain density, is a critical parameter in formation evaluation, such as in calculating formation porosity using density logs (Schlumberger, 1983),

$$\phi = \frac{\rho_{ma} - \rho_b}{\rho_{ma} - \rho_f} \tag{1}$$

Where ϕ is formation porosity, ρ_{ma} matrix density (g/cc), ρ_b formation bulk density (g/cc), ρ_f density of the fluid saturating the rock formation (g/cc).

While bulk density is the direct density-log measurement of the formation, the grain density is usually set at certain values, depending on the lithology of formation (Ma et al., 2002), i.e. 2.65 g/cc, 2.71 g/cc and 2.85 g/cc, for sandstone, limestone and dolostone formations, respectively. Obtaining those set default values in the laboratory core analysis is not straightforward, however, as has documented numerous times in the literature (Ma et al., 2002; Ma et al., 2013; Ma and Amabeoku, 2015; Abdallah et al, 2021).

It is obvious that grain density is affected by several factors including formation rock type and its mineralogical composition (Schon, 2011). In log interpretation, matrix grain density can be estimated by using formation bulk density vs. neutron porosity log cross plot. However, grain density is normally measured in the laboratory, where several methods can be used, such as Boyle's law porosimeter, liquid or helium displacement (Brooks, 1957), or calculated from measured dry weight and grain volume (Dutch et al., 1995).

In addition, grain density may also be calculated from mineralogy (Abdallah et al, 2021), determined by such as X-Ray Diffraction (XRD) or Fourier transform infrared spectroscopy (FTIR), both use crushed powder samples. It has been reported that the powder grain density can be higher than that measured on plugs by Boyle's law porosimeter (Ma et al., 2002). Part of the reasons for this discrepancy has been attributed to core sample cleaning (Singer et al., 2022; Qatari et al., 2024).

This study aims to provide answers to the above observed discrepancies through a systematic experimental test using outcrop samples of carbonate rocks. Challenges and solutions in characterizing clastics rocks, especially shaly sand (Al-Ofi et al., 2024), is beyond the scope of this study.

Experiments

Core Material

24 outcrop carbonate rock samples, from 7 different sources, were selected for the study. They are Indiana Limestone (IND), Silurian Dolomite (SD), Edwards Yellow (EDY) Limestone, Austin Chalk (AC), Guelph Dolomite (GD), Edwards White (EW) Limestone, and Wisconsin (WN) Limestone. Table 1 shows that the helium porosity of these samples ranges from as low as 7% for the tight WN limestone to as high as 30% for the porous AC chalk.

Table 1—Porosity and Rock Type of 24 samples from 7 carbonate outcrops

Batch 1 Samples (T_{drying}=110°C)	Porosity [%]	XRD/FTIR Sample	Batch 2 Samples (T_{drying}=190°C)	Porosity [%]	XRD/FTIR Sample	Rock Type
IND1	16.64	XRD/FTIR	IND7	16.52	XRD/FTIR	Indiana Limestone
IND2	15.15	XRD/FTIR	IND8	16.54		Indiana Limestone
IND3	15.16		IND9	16.88		Indiana Limestone
IND4	18.71		IND10	19.95		Indiana Limestone
IND5	19.09	XRD/FTIR	IND11	18.10	XRD/FTIR	Indiana Limestone
IND6	17.68		IND12	14.48		Indiana Limestone
SD1	17.07		SD2	15.22		Silurian Dolomite

Batch 1 Samples (T_{drying}=110°C)	Porosity [%]	XRD/FTIR Sample	Batch 2 Samples (T_{drying}=190°C)	Porosity [%]	XRD/FTIR Sample	Rock Type
EDY1	24.99	XRD/FTIR	EDY2	24.95		Edwards Yellow Limestone
AC1	29.90		AC2	31.15		Austin Chalk
GD1	8.45		GD2	9.58		Guelph Dolomite
EW1	16.10	XRD/FTIR	EW2	15.98		Edwards White Limestone
WN1	9.54		WN2	6.86		Wisconsin Limestone

Experimental Workflow

As shown in Table 1 and Fig 1, the 24 samples were divided into two batches of 12 samples each (Table 1). The Batch 1samples underwent drying at 110°C, as in routine core analysis, and Batch 2 samples were dried at an elevated temperature of 190°C, while ensuring that rock integrities were not destroyed by heating (Ma and Morrow, 1994). The weight of the samples was measured, and the grain density was computed before and after drying, using helium porosimeter. Then, 7 samples (IND, EDY, EDW) were selected for grain density estimation from XRD and FTIR measurements. Rock powders were prepared by crushing small rock fragments using a Retsch Jaw Crusher BB50, in order to achieve the fineness of powder sample approximately 0.5 mm. The powder was then subjected to milling to an even finer micro scale using a Retsch Oscillating Mixing Mill MM400. The powder obtained was then sieved and collected for particles sizes 63.5 µm and below, then XRD and FTIR tests were performed.

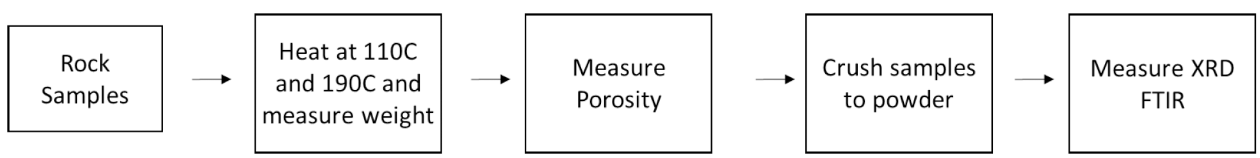

Figure 1—Experimental procedure

Grain Density Measurements

Grain density was calculated by dividing the sample dry weight by its grain volume, measured by a Boyle's Law based helium porosimeter (Temco HP-401), with procedures as described in Abdallah et al (2021). The sample is placed in a chamber of known volume, while helium is held in a reference chamber at known volume and pressure. The two chambers are connected, causing the helium pressure to drop as it fills the sample chamber and the sample pore volume. With measured bulk volume, the grain volume is then derived.

XRD Measurements

An Olympus BTX II benchtop energy dispersive XRD system, operated at 30 kV and 0.370 mA, was used to analyze the sample mineralogy. It was run with an incident angle (2θ) spanning from 5° to 55° with a step size of 0.05°. Stacking the exposure allows maximizing the measurement signal-to-noise ratio (45 exposures were used in this study). The measurement spectrums were processed by a proprietary software, that removes the high and low frequency noise and fits the main and secondary peaks for each of the minerals with the database mineral spectrums. The resultant curve is manually matched with standard spectrum curves of the various minerals. The secondary peaks are used to confirm that the primary peaks actually belong to a mineral (Al-Ofi and Ma, 2023).

FTIR Measurements

FTIR has proven its potential to be used for minerals identification and quantification, as it has characteristic absorption bands in the mid-range of the infrared from 4000 to 400 cm^{-1} (Ruessink and Harville, 1992). In this study. The FTIR measurements were conducted using a Bruker Vertex 80 spectrometer. The FTIR pellets were prepared as per the procedures listed below (Herron et al., 1987; Herron et al., 1998; Abo, 2010).

1. First, 1 g of rock powder was mixed with 10 ml of isopropanol, and then the mixture was micronized for 30 minutes to reduce particle size to <2 μm.
2. The sample (powder and isopropanol) was transferred into a Teflon dish and placed into a vacuum oven set to 110°C for 20 minutes to dry out the isopropanol.
3. 1 mg of the dried powder sample was added to 900 mg of Potassium bromide (KBr) and mixed using Retsch MM 400 Mixer Mill to make a homogenous mixture. Note that KBr was used since it does not show any absorption spectrum in the IR region of interest
4. Finally, 200 mg from the KBr mixtures were loaded into a hydraulic laboratory pellet press system (3636 X-Press®) and pressed at a pressure of 10-ton with 13 mm diameter evacuable die for 10 minutes.

Once the pellets were created, they were placed in the FTIR instrument and measurements were performed.

Results and Discussion

Grain Density from Porosimeter

From the measurement results (Table 2), a range of grain density is observed. Despite the variety of mineralogy (calcites and dolomites), the measured grain density is again lower than the expected often defaulted values, especially for the IND calcite samples. To ensure that the discrepancy is not from the presence of residual water in the samples after conventional drying, two separate batches of samples underwent additional vacuum drying at temperatures of 110°C and 190°C, respectively. The drying was done over a period of 50 hours, and the weight of the samples was taken at different points during that period.

Table 2—Dry weight, grain volume, and calculated grain density of the 24 plug samples in two batches

Batch 1 Samples (110°C)	Weight [g]	Grain Volume [cc]	Grain Density [g/cc]		Batch 2 Samples (190°C)	Weight [g]	Grain Volume [cc]	Grain Density [g/cc]
IND1	122.977	45.973	2.675		IND7	123.912	46.233	2.680
IND2	124.532	46.782	2.662		IND8	123.080	45.841	2.685
IND3	124.906	46.821	2.668		IND9	123.614	46.225	2.674
IND4	116.958	43.809	2.670		IND10	115.936	43.437	2.668
IND5	115.936	43.437	2.669		IND11	117.736	44.091	2.670
IND6	120.452	45.087	2.672		IND12	111.474	41.641	2.677
SD1	130.932	46.706	2.803		SD2	133.954	47.723	2.807
EDY1	111.912	41.380	2.704		EDY2	110.443	40.861	2.703
AC1	101.547	37.698	2.694		AC2	99.349	36.844	2.696
GD1	145.189	51.305	2.830		GD2	145.113	51.247	2.832
EW1	121.522	45.011	2.700		EW2	119.444	44.208	2.702
WN1	141.030	50.763	2.778		WN2	137.583	49.514	2.779

For Batch 1 samples vacuum dried at 110°C, the weight of the samples was taken at the start, and also at 16, 20, 36, and 50 hours of drying. Before each measurement, the samples were given approximately 6 hours to cool down inside the vacuum oven (Table 3). Results show little change in weight reduction. The sample grain density was then calculated after the last drying step at 50 hours, and the results are compared to the initial values (Table 3). The maximum change observed in grain density was only 0.22%.

Table 3—Weight measurements of Batch 1 plug samples dried at 110°C for 50 hours.

Sample	Weight at 0 hr [g]	Grain Density_0 hr [g/cc]	Weight at 16 hr [g]	Weight at 20 hr [g]	Weight at 36 hr [g]	Weight at 50 hr [g]	Grain Density_50 hr [g/cc]	Δ Grain Density [%]
IND1	122.998	2.675	122.969	122.966	122.965	122.965	2.675	0.00
IND2	124.556	2.662	124.528	124.526	124.525	124.525	2.663	0.04
IND3	124.929	2.668	124.901	124.899	124.898	124.898	2.670	0.07
IND4	116.980	2.670	116.929	116.925	116.922	116.921	2.669	-0.04
IND5	115.959	2.669	115.903	115.900	115.899	115.898	2.675	0.22
IND6	120.471	2.672	120.438	120.435	120.434	120.434	2.670	-0.07
SD1	130.924	2.803	130.921	130.919	130.917	130.916	2.808	0.18
EDY1	111.905	2.704	111.898	111.895	111.892	111.892	2.698	-0.22
AC1	101.542	2.694	101.520	101.510	101.502	101.500	2.694	0.00
GD1	145.180	2.830	145.158	145.142	145.139	145.140	2.831	0.04
EW1	121.520	2.700	121.502	121.491	121.489	121.490	2.698	-0.07
WN1	141.026	2.778	141.010	140.998	140.997	140.997	2.778	0.00

Similar procedures were followed for Batch 2 plug samples, dried at 190°C. Due to the higher temperature, the cooling time before weight measurements was increased to 9 hours. The weight was measured at the start and after 16, 32, and 50 hrs of drying (Table 4). Here again, the change in weight after drying at a higher temperature is insignificant, although slightly higher than those Batch 1 plug samples that dried at 110°C. The grain density of the samples was also computed after the last drying step, and the results are summarized in Table 4. The maximum change observed in the grain density was 0.285%, again not significant. These results confirm that, despite efforts to dry the samples under an elevated temperature, the grain densities are still not aligned with the expected values.

Table 4—Weight measurements of Batch 2 plug samples dried at 190°C for 50 hours.

Sample	Weight at 0 hr [g]	Grain Density_0 hr [g/cc]	Weight at 16 hr [g]	Weight at 32 hr [g]	Weight at 50 hr [g]	Grain Density_50 hr [g/cc]	Δ Grain Density [%]
IND1	123.938	2.680	123.858	123.857	123.842	2.686	0.224
IND2	123.105	2.685	123.016	123.014	123.001	2.683	-0.074
IND3	123.633	2.674	123.566	123.564	123.562	2.678	0.150
IND4	113.437	2.668	113.344	113.328	113.214	2.669	0.037
IND5	117.754	2.670	117.678	117.677	117.674	2.673	0.112
IND6	111.482	2.677	111.434	111.432	111.431	2.683	0.224
SD1	133.949	2.807	133.649	133.644	133.640	2.799	-0.285
EDY1	110.438	2.703	110.418	110.417	110.416	2.706	0.111
AC1	99.350	2.696	99.308	99.308	99.305	2.701	0.185

Sample	Weight at 0 hr [g]	Grain Density_0 hr [g/cc]	Weight at 16 hr [g]	Weight at 32 hr [g]	Weight at 50 hr [g]	Grain Density_50 hr [g/cc]	Δ Grain Density [%]
GD1	145.110	2.832	144.995	144.993	144.989	2.834	0.071
EW1	119.444	2.702	119.402	119.401	119.399	2.700	-0.074
WN1	137.589	2.779	137.543	137.542	137.540	2.778	-0.036

Grain Density from Mineralogy

XRD mineralogy measurements were performed on 7 calcite samples (Table 1). Two samples that showed grain densities similar to the default value of ≈2.71 g/cc were also included in this study (Tables 2, 3, and 4). Prior to making the measurements, the powder was dried at 100°C for 24 hours to remove all traces of moisture. The acquired spectra, as exampled in Figure 2, were analyzed by the commercial software, Xpowder, to provide the specific composition of the samples.

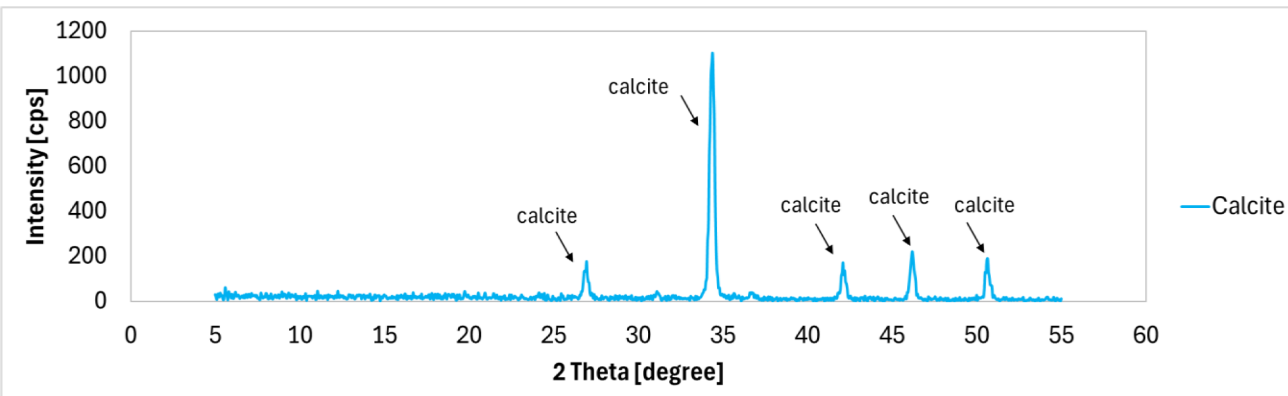

Figure 2—XRD spectra for powder sample IND5.

Using the obtained mineralogy and densities for pure minerals (Schlumberger log interpretation chartbook, 1997), the grain density can be calculated following the mixing rule (Eberl, 2003; Abdallah et al, 2021).

$$\rho_{ma} = \sum_i v_i \rho_i \tag{2}$$

Where ρ_i is the individual mineral grain density and v_i is the volumetric fraction of that mineral.

The XRD spectra of all the samples tested revealed that the main mineral is calcite. For the Indiana limestone, up to 7.3% quartz was observed in some of the samples (Table 5).

Table 5—XRD mineralogy and grain density of 7 selected limestone samples.

Powder Sample	XRD			FTIR				
	Calcite	Quartz	Cal. Grain density	Calcite	Quartz	Kaolinite	Montmorillonite	Cal. Grain density
	%	%	g/cc	%	%	%	%	g/cc
IND1	97.6	2.4	2.709	95.17	2.63	0.57	1.64	2.698
IND2	97.3	2.7	2.698	96.42	0	0	3.58	2.689
IND5	100	0	2.710	96.64	0.66	0	2.69	2.693
IND7	97.3	2.7	2.708	98.65	0.93	0	0.42	2.707
IND11	92.7	7.3	2.710	93.80	3.88	0	2.31	2.693

Powder Sample	XRD			FTIR				
	Calcite	Quartz	Cal. Grain density	Calcite	Quartz	Kaolinite	Montmorillonite	Cal. Grain density
	%	%	g/cc	%	%	%	%	g/cc
EDY-1	100	0	2.710	100	0	0	0	2.710
EW-1	100	0	2.710	100	0	0	0	2.710

The powder samples were also prepared for FTIR measurements to further quality control the XRD mineralogy, for that it has been reported that FTIR can provide more detailed information about minerals than that of XRD (Craddock et al., 2017). As for the XRD test (Al-Ofi and Ma, 2023), one of the critical steps for a FTIR test is also sample preparation (Popko, 1999). The carefully prepared pellets were loaded into the FTIR instrument and their spectra were measured and analyzed. In the example of Figure 3, the baseline FTIR spectrum of a pure KBr sample is shown in black, while the sample spectrum in blue. Results indicate that the FTIR test does generally provide a more detailed information about mineralogy than XRD. In particular, the presence of some clay minerals was identified (Table 5). It is interesting to note that for samples IND2 and 5, XRD and FTIR results show controversies in the presence of quartz, illustrating the solution non-uniqueness in the process of mineral identification from either XRD or FTIR (Al-Ofi and Ma, 2023). XRF elemental analysis of Si would have been helpful to resolve this discrepancy.

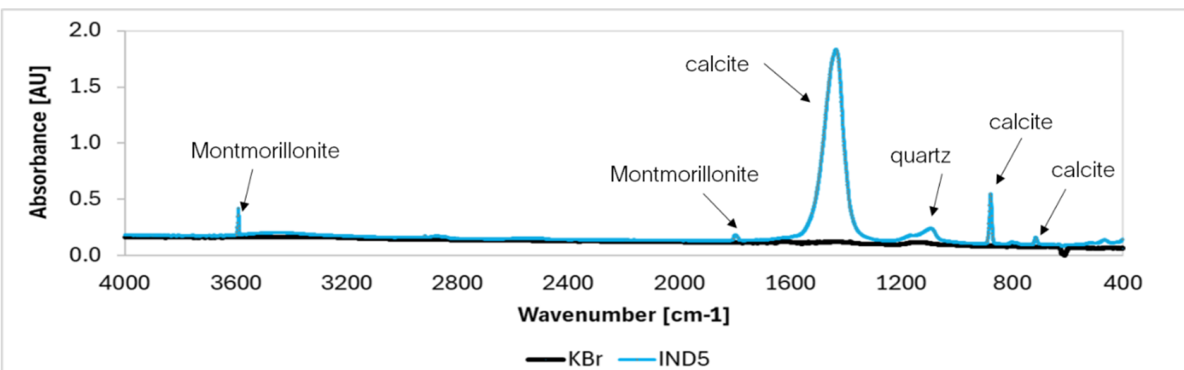

Figure 3—FTIR spectra for powder sample IND5

With FTIR mineralogy determined, the grain density was calculated in a similar manner to that of the XRD measurements. The densities of Kaolinite and Montmorillonite (2.62 g/cc and 2.12 g/cc, respectively) were also taken from the Schlumberger chartbook (Schlumberger log interpretation chartbook, 1997). The calculated grain densities are shown in Table 5.

Some differences between the densities computed from XRD and FTIR measurements were observed as shown in Figure 4, due to the presence of clays detected by FTIR, though the difference is small due to the concentration of clays is low, <5%. However, existing of clays does reduce the matrix density for the clay minerals are lighter than calcite.

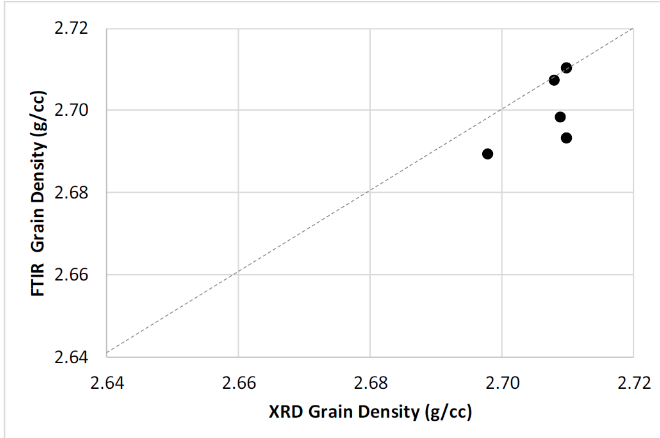

Figure 4—Grain density results comparison between FTIR and XRD measurements.

Beside mineralogy impurity, Indiana limestone is also known containing micro-porosity (SayedAkram et al, 2016). Some of these porosities might be inaccessible and/or isolated (Jin et al., 2022) as were shown by the SEM images taken by Arns et al (2019). If only 1 p.u. of the total porosity were isolated, the calculated grain density would drop from 2.710 g/cc to 2.670 g/cc, a significant factor to consider when interpreting measured plug grain density.

To evaluate micro/isolated porosity, thin section and SEM images of the Indiana limestone samples that were studied before (Al-Hamad et al., 2024) were analyzed. Figure 5 represents a magnified version of a thin section image, which clearly show that the major component of the porosity is intergranular, with meaningful amount of micro-porosity. In addition, a clear cement extension is visible between the grains and the macro porosity. This cement may render some of the macro-porosity inaccessible as the fluid pathways may not be connected to the outer boundaries of the sample.

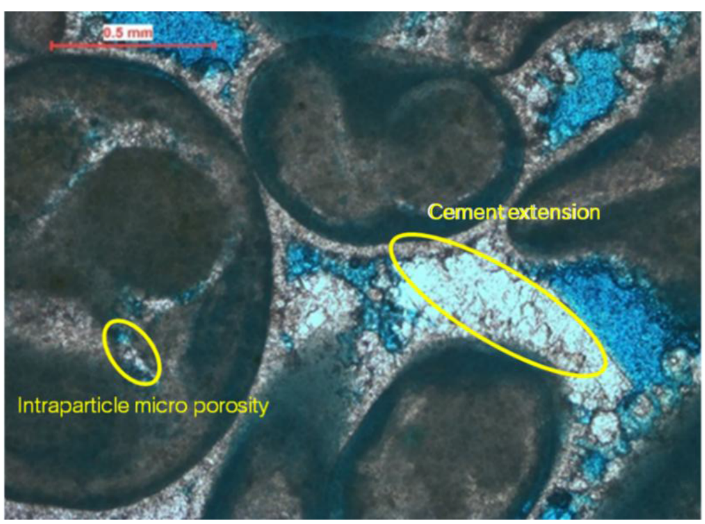

Figure 5—Magnified thin section images of Indiana limestone.

These observations were also confirmed by the high resolution SEM image taken for the same sample (Figure 6), which was readjusted to black and white colors to better illustrate the pore structure. From the SEM image, clear micro porosities are observed between the grains. Further studies would be required to fully understand and identify the isolated porosity (Jin et al., 2022) and its implications on grain density measurements.

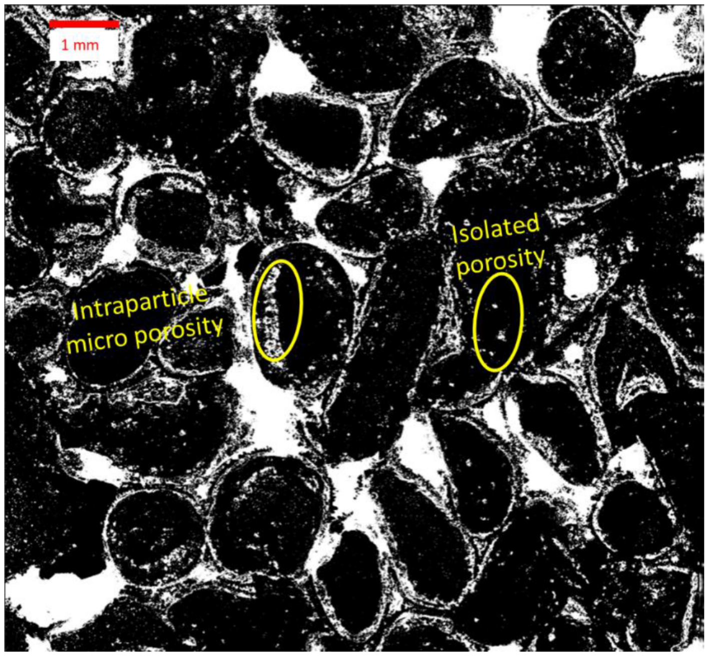

Figure 6—SEM image of Indiana limestone.

Based on these results and observations, it can be concluded that the often observed difference between the measured and commonly defaulted grain density of carbonate rocks may be due to mainly the below two factors, as illustrated in Figure 7.

- Mineral impurity. Existing of minor amount of other minerals such as clays will reduced measured matrix density for clay minerals are lighter than carbonate minerals.
- Isolated/inaccessible porosity. Content in isolated/inaccessible pores is water, and it will reduce measured grain density if isolated porosity exists in a meaningful amount.

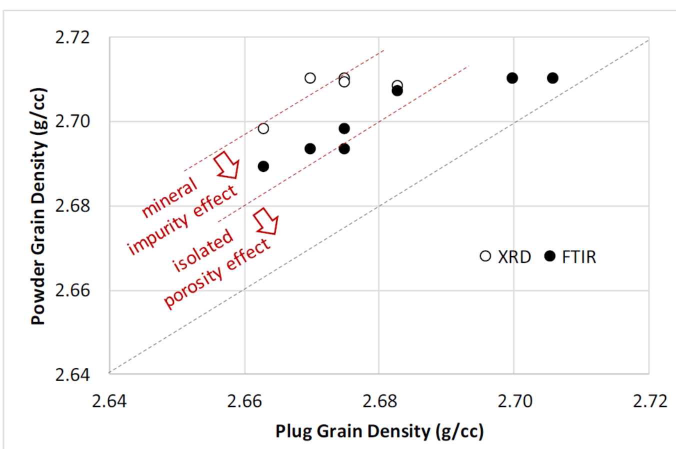

Figure 7—Comparison between plug grain density and that derived from FTIR/XRD powder measurements.

Conclusion

The lower than expected matrix grain density value from laboratory core analysis for carbonate rocks was systematically investigated in this study, covering effect of drying, mineral impurity, and isolated porosity.

Effect of drying. Carbonate samples were dried at 110°C and 190°C, and effect of drying conditions on matrix density measurement is insignificant.

Mineral impurity. XRD test indicates that the Indianan limestone is basically calcite with some quartz content, while FTIR results indicate presence of clays. Both quartz and clays will reduce matrix density.

Isolated porosity. Examining images of thin sections and SEM shows that the Indiana limestone contains a meaningful amount of isolated porosity which may not be accessible. Its presence would reduce matrix density.

To further study isolated porosity effects, image based digital rock physics may be hvaluable. Multifrequency dielectric dispersion studies may also help to better understand existence of isolated porosity in carbonate rocks.

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